

INNOMINE PRODUCT REQUIREMENTS DOCUMENT (PRD)

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Project Name: Innomine

Category: Innovation-Focused E-commerce Marketplace

Prepared By: Mohammed Kazim Khan & OpenAI ChatGPT

1. Executive Summary

1.1 Introduction

Innomine is a next-generation digital marketplace dedicated exclusively to innovative physical products. Unlike conventional e-commerce platforms that primarily serve mass-market consumer goods, Innomine provides a specialized ecosystem where inventors, engineers, researchers, startups, manufacturers, student innovators, and technology companies can showcase, market, and sell innovative products to a global audience.

The platform addresses one of the biggest challenges faced by innovators: product discovery. Many founders develop technically impressive products but struggle to reach customers because they rely on standalone websites with limited visibility and expensive marketing campaigns. Existing marketplaces prioritize established brands and high-volume products, making it difficult for emerging innovations to gain attention.

Innomine bridges this gap by creating a centralized marketplace where innovation is the primary focus. Beyond standard e-commerce functionality, the platform integrates modern content-driven discovery through short-form videos, founder profiles, intelligent search, community engagement, and advanced recommendation systems. The goal is to transform product discovery from a traditional search-based experience into an engaging exploration of new ideas and technologies.

The long-term vision is to establish Innomine as the world's leading destination for innovative products, enabling creators to focus on innovation while the platform provides visibility, trust, and access to customers.

2. Vision Statement

To become the world's most trusted marketplace dedicated exclusively to innovative physical products, empowering creators to transform ideas into globally recognized products through technology, community, and commerce.

3. Mission Statement

Innomine exists to remove the barriers that prevent innovative products from reaching customers. By providing a premium marketplace with integrated marketing, product storytelling, and advanced commerce tools, the platform enables innovators to focus on building exceptional products while Innomine simplifies discovery, trust, sales, and customer engagement.

4. Problem Statement

The current innovation ecosystem suffers from several structural challenges that limit the commercial success of innovative products.

Many inventors, startups, university research groups, and hardware companies build exceptional products but lack the resources to market them effectively. Their products are commonly hosted on standalone websites that receive minimal organic traffic, forcing founders to spend heavily on digital advertising or rely on crowdfunding campaigns with limited visibility.

Traditional marketplaces such as Amazon and Flipkart are optimized for high-volume retail products rather than showcasing innovation. Their recommendation systems generally favor products with established sales histories, making it difficult for newly launched innovations to gain exposure. As a result, groundbreaking products often remain undiscovered despite solving meaningful real-world problems.

Customers interested in discovering innovative products also face challenges. There is no dedicated marketplace where they can confidently explore emerging technologies, compare inventions, learn about creators, and purchase innovative products from verified founders. Product pages frequently rely on static images and text, which are often insufficient to communicate the value and functionality of novel products.

These challenges create a fragmented innovation ecosystem in which creators struggle to reach buyers, and buyers struggle to discover meaningful innovations.

5. Proposed Solution

Innomine is designed as a specialized digital marketplace focused exclusively on innovation. The platform combines premium e-commerce functionality with modern content-driven discovery to create an ecosystem where innovative products can thrive.

The marketplace enables innovators to establish verified storefronts, publish detailed product listings, upload demonstration videos, manage inventory, process orders, and interact directly with customers. Buyers can discover products through intelligent search, curated recommendations, personalized feeds, and an immersive short-form video experience that highlights product functionality and founder stories.

Rather than requiring innovators to build independent websites and generate their own traffic, Innomine centralizes product discovery, allowing customers interested in innovation to explore products from multiple creators within a single trusted platform.

6. Market Opportunity

The global innovation economy continues to expand due to increasing investment in technology startups, hardware manufacturing, robotics, IoT, AI-enabled devices, renewable energy solutions, healthcare technologies, educational products, and consumer electronics.

Several factors create a significant opportunity for Innomine:

- Rapid growth of startup ecosystems.
- Increasing university innovation programs.
- Expansion of maker communities.
- Growth of crowdfunding platforms.
- Rising demand for unique and problem-solving products.
- Increased interest in sustainable and smart technologies.
- Wider adoption of digital commerce worldwide.

Despite these trends, no major global marketplace is dedicated exclusively to innovative physical products. This presents an opportunity for Innomine to become the preferred platform for innovation-driven commerce.

7. Value Proposition

Value for Buyers

- Access to products unavailable on traditional marketplaces.
- Discovery of emerging technologies from verified innovators.
- Rich product demonstrations through videos and interactive content.
- Trusted purchasing experience backed by platform verification.
- Opportunity to support startups, inventors, and engineering communities.

Value for Innovators

- Immediate exposure to a targeted audience interested in innovation.
 - Reduced customer acquisition costs compared to operating independent websites.
 - Professional storefront creation without technical expertise.
 - Integrated marketing through the platform's recommendation systems and discovery feed.
 - Performance analytics, sales tracking, and customer insights.
 - Community engagement through followers, reels, and creator branding.
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8. Unique Selling Proposition (USP)

Innomine differentiates itself by combining commerce with creator-driven content.

Every product is supported by an interactive ecosystem including:

- Product demonstration reels.
- Founder stories.

- Development journey videos.
- Manufacturing process clips.
- Product tutorials.
- Customer demonstrations.
- Innovation highlights.

Innomine differentiates itself through three core pillars:

- **Innovation Marketplace** – A marketplace exclusively dedicated to innovative physical products.
 - **Innovation Feed** – A short-form video discovery experience where every reel is directly linked to a product.
 - **Innovation Verification System** – An optional trust framework that allows innovators to verify their products through patents, research, awards, certifications, or other supporting evidence, helping buyers identify authentic innovations.
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9. Product Principles

Every decision made during the design and development of Innomine must follow these principles.

Innovation First

Every feature should promote innovative products rather than generic retail items.

Creator Empowerment

The platform exists to help innovators succeed. Features should simplify selling, marketing, and customer engagement.

Premium User Experience

Every page should feel modern, elegant, responsive, and intuitive.

Trust and Transparency

Verified creators, authentic reviews, secure payments, and transparent policies are fundamental to the platform.

Simplicity

Despite offering advanced functionality, the user experience should remain simple and easy to understand.

Performance

Fast loading times, smooth animations, responsive interactions, and optimized media are mandatory.

Accessibility

The platform should follow modern accessibility standards and remain usable across devices and user needs.

10. Success Metrics (KPIs)

The success of Innomine will be measured using quantifiable business and technical metrics.

Business KPIs

- Monthly Active Users (MAU)
- Daily Active Users (DAU)
- Total Registered Buyers
- Total Verified Innovators
- Products Listed
- Orders Completed
- Gross Merchandise Value (GMV)
- Average Order Value
- Seller Retention Rate
- Buyer Retention Rate
- Platform Revenue
- Conversion Rate
- Customer Acquisition Cost
- Lifetime Customer Value

Engagement KPIs

- Reel Views
- Average Watch Time
- Products Shared
- Products Saved
- Likes and Comments
- Repeat Purchases
- Session Duration
- Search Success Rate

Technical KPIs

- Average Page Load Time

- API Response Time
 - Uptime Percentage
 - Lighthouse Performance Score
 - Accessibility Score
 - SEO Score
 - Crash Rate
 - Error Rate
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11. Target Audience

Innomine is designed to serve a diverse ecosystem centered around innovation. Rather than targeting general online shoppers, the platform focuses on individuals and organizations actively involved in creating, discovering, evaluating, purchasing, or investing in innovative products.

Primary Audience

Innovators

Individuals or teams who have developed innovative physical products and require a professional platform to showcase, market, and sell them.

Examples include:

- Startup founders
 - Engineers
 - Student innovators
 - Independent inventors
 - Researchers
 - Product designers
 - Hardware developers
 - Robotics teams
 - IoT developers
 - Makers
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Buyers

Customers interested in discovering products that solve real-world problems through innovation.

This audience includes:

- Technology enthusiasts
 - Early adopters
 - Professionals
 - Hobbyists
 - Businesses
 - Educational institutions
 - Research organizations
 - Government organizations
 - Manufacturing companies
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Secondary Audience

- Investors searching for promising startups.
 - Incubators and accelerators.
 - University innovation labs.
 - Innovation competitions.
 - Engineering colleges.
 - Technology communities.
 - Media organizations.
 - Corporate innovation teams.
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12. User Personas

To ensure that every feature serves real users, Innomine is designed around several representative personas.

Persona 1 – The Innovator

Name: Arjun Rao

Age: 26

Occupation: Startup Founder

Goals

- Launch innovative products.

- Increase visibility.
- Receive customer feedback.
- Generate sales.
- Build a recognizable brand.

Pain Points

- Expensive marketing.
- Low website traffic.
- Difficulty building customer trust.
- Poor product discoverability.

Needs

- Easy product listing.
- Analytics dashboard.
- Store customization.
- Video-based marketing.
- Verified seller badge.
- Order management.

Persona 2 – The Buyer

Name: Sarah Wilson

Age: 31

Occupation: Software Engineer

Goals

- Discover innovative products.
- Purchase unique technology.
- Learn about founders.
- Compare products.
- Receive trusted recommendations.

Pain Points

- Difficult to find genuine innovations.
- Static product descriptions.
- Lack of demonstrations.

- Fake reviews.

Needs

- Rich product pages.
 - Demo videos.
 - Verified innovators.
 - Wishlist.
 - Product comparison.
 - Personalized recommendations.
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Persona 3 – The Investor

Name: David Chen

Age: 39

Occupation: Angel Investor

Goals

- Identify promising startups.
- Evaluate products.
- Contact founders.
- Track emerging technologies.

Needs

- Founder profiles.
 - Product traction metrics.
 - Innovation categories.
 - Direct communication.
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Persona 4 – Platform Administrator

Responsible for maintaining trust and quality across the platform.

Responsibilities include:

- Product moderation.
- Seller verification.
- Fraud prevention.
- User management.

- Analytics.
 - Category management.
 - Platform health monitoring.
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13. User Roles and Permissions

The platform follows Role-Based Access Control (RBAC).

Guest User

Permissions:

- Browse products.
- Search.
- Watch reels.
- View innovator profiles.
- Read blogs.
- Explore categories.

Restrictions:

- Cannot purchase.
 - Cannot comment.
 - Cannot upload.
 - Cannot create wishlists.
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Registered Buyer

Permissions:

- Purchase products.
- Wishlist.
- Cart.
- Follow innovators.
- Like reels.
- Comment.
- Rate products.
- Share products.

- Track orders.
 - Save reels.
 - Receive notifications.
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Verified Innovator (Seller)

Permissions:

- Everything available to buyers.
 - Create storefront.
 - Upload products.
 - Upload reels.
 - Manage inventory.
 - View analytics.
 - Process orders.
 - Respond to reviews.
 - Create discounts.
 - Access sales dashboard.
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Moderator

Permissions:

- Review reported content.
 - Remove harmful posts.
 - Approve products.
 - Moderate reels.
 - Moderate comments.
-

Administrator

Full platform access.

Capabilities include:

- User management.
- Platform analytics.
- Seller verification.

- Financial reports.
 - CMS management.
 - Platform configuration.
 - Category management.
 - Coupon management.
 - Feature management.
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14. Business Model

Innomine follows a multi-revenue marketplace model.

Instead of depending on a single revenue source, the platform generates sustainable income through multiple channels.

Marketplace Commission

A percentage of every successful sale.

Premium Innovator Plans

Subscription plans providing:

- Better visibility.
 - Premium storefronts.
 - Enhanced analytics.
 - Advanced marketing tools.
 - Priority customer support.
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Featured Product Promotions

Innovators can promote products on:

- Home page.
 - Trending section.
 - Category pages.
 - Innovation feed.
 - Search results.
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Sponsored Innovation Campaigns

Companies can sponsor innovation campaigns or themed showcases.

Platform Advertising

Limited, relevant advertising for innovation-focused brands and partners.

Enterprise Partnerships

Partnerships with:

- Universities.
 - Research organizations.
 - Incubators.
 - Accelerators.
 - Corporate innovation teams.
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15. Revenue Model

The long-term financial strategy combines transactional, subscription, promotional, and partnership revenue.

Primary revenue sources:

- Marketplace commission.
- Premium subscriptions.
- Featured listings.
- Sponsored campaigns.
- Advertising.
- Enterprise partnerships.

Future revenue sources:

- API services.
 - AI-powered analytics.
 - SaaS tools for innovators.
 - Logistics partnerships.
 - Product certification services.
 - Innovation awards and events.
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16. Complete Feature Inventory

This section defines every major capability planned for Innomine.

Public Website

- Landing page
- About
- Innovation categories
- Product catalog
- Product details
- Search
- Advanced filters
- Trending innovations
- Featured innovators
- Innovation feed
- Blog
- Careers
- Contact
- FAQs
- Terms
- Privacy

Authentication

- Register
- Login
- Email verification
- Forgot password
- Reset password
- Google sign-in
- GitHub sign-in
- Two-factor authentication (future)

Buyer Features

- User profile
 - Shopping cart
 - Wishlist
 - Order tracking
 - Saved reels
 - Notifications
 - Reviews
 - Ratings
 - Follow innovators
 - Personalized recommendations
 - Purchase history
 - Address management
 - Payment history
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Innovator Features

- Seller onboarding
 - Verification workflow
 - Store profile
 - Product management
 - Inventory management
 - Reel uploads
 - Sales dashboard
 - Revenue analytics
 - Customer insights
 - Order management
 - Coupons
 - Store customization
 - Team management (future)
-

Reels & Community

The reels system is a **core product**, not a marketing add-on.

Features include:

- Vertical short-form videos
- Product tagging
- Buy Now CTA
- Like
- Comment
- Share
- Save
- Follow creator
- Report content
- Auto-play feed
- Creator profiles
- Related products
- Trending feed
- Category feed
- Personalized recommendations
- Founder story series
- Product launch announcements

Innovation Verification System

Features:

- Product verification request
- Patent verification
- Research paper submission
- Award verification
- Prototype verification
- Sustainability verification
- Government certification support
- Innovation badges
- Trust score
- Verification review workflow

- Verification status tracking

E-commerce

- Product catalog
- Cart
- Checkout
- Coupons
- Secure payments
- Order history
- Refund requests
- Returns management
- Shipping tracking
- Tax calculation
- Invoice generation

Admin Features

- Dashboard
 - User management
 - Seller verification
 - Product moderation
 - Reel moderation
 - Analytics
 - Reports
 - Revenue management
 - Category management
 - CMS
 - Notification management
 - Feature flags
 - Platform monitoring
-

Future Features

- AI recommendations
 - AI chatbot
 - Live shopping
 - AR product previews
 - Product crowdfunding
 - Investor networking
 - Innovation competitions
 - B2B marketplace
 - Mobile applications
 - Creator monetization
 - Affiliate program
 - International shipping
 - Multi-language support
 - Carbon footprint indicators
 - Patent/reference links (optional for innovators)
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17. Functional Requirements

17.1 User Authentication & Authorization

The platform shall provide a secure authentication system supporting multiple user roles while maintaining high security and usability.

Registration

The system shall allow users to register using:

- Email & Password
- Google OAuth
- GitHub OAuth
- (Future) Apple Sign-In

Each user must select an account type:

- Buyer
- Innovator (Seller)

The registration process shall include:

- Email verification
 - Password strength validation
 - CAPTCHA verification
 - Acceptance of Terms & Privacy Policy
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Login

The platform shall support:

- Secure login
 - Remember Me
 - Multi-device sessions
 - Session expiration
 - Automatic logout after inactivity
 - Secure JWT authentication
 - Refresh Token mechanism
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Authorization

The platform shall implement Role-Based Access Control (RBAC).

Roles include:

- Guest
- Buyer
- Innovator
- Moderator
- Administrator

Each role shall only access permitted resources.

17.2 User Profile Management

Every registered user shall have a profile.

The profile shall support:

- Personal Information
- Profile Picture
- Contact Information

- Shipping Addresses
 - Saved Payment Methods
 - Notification Preferences
 - Account Privacy
 - Security Settings
 - Password Change
 - Account Deletion
-

17.3 Product Management

Innovators shall be able to:

- Create Products
- Edit Products
- Archive Products
- Delete Draft Products
- Manage Inventory
- Upload Images
- Upload Videos
- Upload Technical Documents
- Add Product Specifications
- Add Warranty Information
- Add Shipping Information
- Set Pricing
- Configure Discounts
- Manage Stock

Products shall support multiple media assets.

17.4 Product Discovery

Users shall discover products through:

- Categories
- Search
- Trending

- Featured Products
- Editor Picks
- Latest Innovations
- Personalized Recommendations
- Innovation Feed
- Related Products

Advanced filtering shall include:

- Category
 - Price
 - Brand
 - Rating
 - Innovation Type
 - Technology
 - Availability
 - Seller
 - Country
-

17.5 Shopping Experience

The platform shall provide:

- Shopping Cart
 - Wishlist
 - Save for Later
 - Product Comparison
 - Coupon System
 - Checkout
 - Tax Calculation
 - Shipping Calculation
 - Invoice Generation
 - Order Confirmation
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17.6 Innovation Feed (Core Platform Feature)

The Innovation Feed is one of the primary discovery systems of Innomine.

Every reel shall support:

- Vertical scrolling
- Auto-play
- Product tagging
- Buy Now button
- Like
- Comment
- Share
- Save
- Follow Innovator
- Visit Product
- Visit Store

Every reel shall contain:

- Product information
- Creator information
- Innovation category
- Product price
- Product thumbnail
- Verification badge (if applicable)

Future enhancements:

- Live Product Launch
- Live Shopping
- AI Generated Captions
- Auto Translation
- Multi-language subtitles

17.7 Innovation Verification System

The Innovation Verification System is a trust-building mechanism that enables innovators to validate the authenticity and credibility of their products.

Participation in the verification program is optional but strongly encouraged.

Innovators may submit supporting documentation, including:

- Patent Numbers
- Patent Pending Documents
- Research Publications
- White Papers
- Technical Documentation
- Prototype Demonstrations
- Competition Awards
- Incubator or Accelerator Certifications
- Government Recognitions
- Sustainability Certifications

The platform shall provide:

- Verification Request Submission
- Document Upload
- Verification Status Tracking
- Admin Review Workflow
- Verification Approval/Rejection
- Resubmission after rejection
- Verification History

Approved products shall display verification badges across:

- Product Pages
- Product Listings
- Innovation Feed
- Search Results
- Seller Profiles

17.8 Order Management

Buyers shall be able to:

- Place Orders
- Track Orders
- Cancel Orders (within policy)

- Download Invoices
- View Order History
- Request Returns
- Request Refunds
- Contact Seller

Innovators shall be able to:

- Accept Orders
 - Process Orders
 - Update Shipping Status
 - Manage Inventory
 - Print Shipping Labels
-

17.9 Payment System

The payment module shall support:

- Secure Checkout
 - Multiple Payment Methods
 - UPI
 - Credit Cards
 - Debit Cards
 - Net Banking
 - Wallets
 - International Payments
 - Payment History
 - Refund Processing
 - Failed Payment Recovery
-

17.10 Notifications

Users shall receive notifications for:

- Order Updates
- Payment Status
- Shipment Updates

- New Followers
- Comments
- Likes
- Product Approval
- Verification Approval
- Promotional Campaigns

Notifications shall support:

- In-App Notifications
- Email Notifications
- Push Notifications (Future)

17.11 Search System

The search engine shall support:

- Keyword Search
- Auto Suggestions
- Search History
- Recent Searches
- Trending Searches
- Category Filters
- Voice Search (Future)
- Image Search (Future)

17.12 Seller Dashboard

The Seller Dashboard shall provide:

- Revenue Overview
- Sales Analytics
- Product Management
- Inventory Management
- Reel Management
- Followers
- Customer Insights

- Orders
 - Reviews
 - Coupons
 - Verification Status
 - Store Settings
-

17.13 Admin Dashboard

Administrators shall have access to:

- Dashboard Analytics
 - User Management
 - Seller Verification
 - Product Moderation
 - Innovation Verification
 - Reel Moderation
 - Reports
 - Payment Monitoring
 - Category Management
 - CMS
 - Support Tickets
 - Audit Logs
 - Platform Configuration
-

17.14 Customer Support

The platform shall provide:

- Help Center
- FAQ
- Contact Support
- Live Chat (Future)
- Ticket Management
- Complaint Resolution
- Seller Support

- Buyer Support
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17.15 Analytics

The platform shall collect analytics for:

- Product Views
 - Reel Views
 - Sales Performance
 - Conversion Rates
 - Customer Retention
 - Seller Growth
 - Search Analytics
 - User Engagement
 - Revenue Trends
-

18. Non-Functional Requirements

While functional requirements define **what the system should do**, non-functional requirements define **how well the system should perform**. These requirements ensure that Innomine delivers a secure, scalable, performant, and reliable experience suitable for production deployment.

18.1 Performance

The platform shall provide a fast and responsive user experience across all supported devices.

Requirements

- Initial page load time shall be less than **2 seconds** on broadband connections.
 - Time to Interactive (TTI) shall be under **3 seconds**.
 - API response time should average **<300 ms** for standard operations.
 - Product search results shall be returned in under **500 ms**.
 - Infinite scrolling shall remain smooth without noticeable frame drops.
 - Images shall use lazy loading and next-generation formats (WebP/AVIF where supported).
 - Videos shall use adaptive streaming for efficient playback.
-

18.2 Scalability

The architecture shall support future growth without requiring significant redesign.

The system shall support:

- Millions of registered users.
 - Millions of products.
 - Millions of uploaded reels.
 - High concurrent user traffic.
 - Horizontal server scaling.
 - Cloud-native deployment.
 - CDN integration.
 - Object storage for media assets.
 - Future migration to microservices if required.
-

18.3 Availability & Reliability

The platform shall be designed for high availability.

Requirements:

- Target uptime: **99.9%**
 - Automatic recovery from application failures.
 - Graceful error handling.
 - Retry mechanisms for transient failures.
 - Scheduled backups.
 - Disaster recovery strategy.
 - Monitoring and alerting.
-

18.4 Security

Security is a core requirement.

The platform shall implement:

Authentication

- JWT Authentication
- Refresh Tokens
- Secure Session Management

- Password Hashing (bcrypt or Argon2)

Authorization

- Role-Based Access Control (RBAC)
- Permission validation
- Route protection
- API authorization

Application Security

- HTTPS enforcement
- Input validation
- Output encoding
- XSS protection
- CSRF protection
- SQL/NoSQL injection protection
- File upload validation
- Rate limiting
- Secure HTTP headers
- CORS configuration

Data Security

- Encrypted passwords
- Secure token storage
- Encrypted sensitive information
- Secure payment handling
- GDPR-ready privacy architecture (where applicable)

18.5 Accessibility

The platform shall comply with modern accessibility practices.

Support:

- Keyboard navigation.
- Screen readers.
- Focus indicators.
- High contrast.

- Color accessibility.
- Alt text.
- Semantic HTML.
- Accessible forms.
- Responsive font scaling.

Target compliance:

WCAG 2.2 AA

18.6 Responsiveness

The platform shall provide a consistent experience across:

- Mobile
- Tablet
- Laptop
- Desktop
- Ultra-wide displays

The application shall follow a **mobile-first design philosophy**.

18.7 Maintainability

The codebase shall be:

- Modular.
 - Reusable.
 - Type-safe.
 - Documented.
 - Easy to test.
 - Easy to extend.
 - Consistent with coding standards.
-

18.8 Compatibility

Supported browsers:

- Chrome
- Edge

- Firefox
- Safari

Supported operating systems:

- Windows
 - macOS
 - Linux
 - Android
 - iOS
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18.9 SEO Requirements

The public website shall support:

- Semantic HTML
 - Structured data (Schema.org)
 - Open Graph
 - Twitter Cards
 - XML Sitemap
 - Robots.txt
 - Canonical URLs
 - Optimized metadata
 - Server-side rendering where appropriate
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18.10 Monitoring & Logging

The platform shall support:

- Error logging
 - Activity logs
 - API monitoring
 - Performance monitoring
 - Security monitoring
 - User analytics
 - Crash reporting
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19. User Journey Maps

This section defines the expected end-to-end experience for each major user type.

19.1 Buyer Journey

1. Discover Innomine through search, social media, referrals, or marketing campaigns.
 2. Browse featured innovations on the landing page.
 3. Explore categories or search for products.
 4. Watch innovation reels and demonstrations.
 5. View detailed product information.
 6. Compare products or save them to a wishlist.
 7. Add products to the cart.
 8. Complete secure checkout.
 9. Track the shipment.
 10. Receive the product.
 11. Submit ratings and reviews.
 12. Follow favorite innovators and continue discovering new products.
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19.2 Innovator Journey

1. Register as an Innovator.
 2. Complete identity verification.
 3. Create a storefront.
 4. Upload products.
 5. Upload reels.
 6. Submit optional innovation verification.
 7. Receive moderation approval.
 8. Publish products.
 9. Receive orders.
 10. Manage inventory.
 11. Analyze sales and customer engagement.
 12. Grow a follower community.
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19.3 Administrator Journey

1. Review new user registrations.
 2. Verify innovators.
 3. Moderate products.
 4. Review innovation verification requests.
 5. Moderate reels and comments.
 6. Monitor platform analytics.
 7. Manage categories and featured products.
 8. Generate reports.
 9. Maintain platform quality and security.
-

20. Information Architecture

The platform shall be organized into the following major modules:

Public Website

- Home
 - About
 - Categories
 - Product Listings
 - Product Details
 - Innovation Feed
 - Innovator Profiles
 - Blog
 - Contact
 - Careers
 - FAQs
 - Legal Pages
-

Buyer Portal

- Dashboard
- Orders
- Wishlist

- Saved Reels
 - Notifications
 - Addresses
 - Payment Methods
 - Profile
 - Settings
-

Innovator Portal

- Dashboard
 - Storefront
 - Products
 - Inventory
 - Orders
 - Reels
 - Analytics
 - Followers
 - Reviews
 - Verification
 - Coupons
 - Settings
-

Administration Portal

- Dashboard
- User Management
- Seller Management
- Innovation Verification
- Product Moderation
- Reel Moderation
- Reports
- Categories
- CMS

- Support
 - Platform Configuration
 - Audit Logs
-

21. Page Inventory & Navigation Structure

This section lists every major page planned for the platform.

Public Pages

- Landing Page
- About
- Categories
- Category Details
- Product Listing
- Product Details
- Innovation Feed
- Innovator Profile
- Blog
- Blog Details
- Contact
- Careers
- FAQ
- Privacy Policy
- Terms & Conditions
- Cookies Policy

Authentication

- Login
- Register
- Forgot Password
- Reset Password
- Email Verification

Buyer

- Dashboard

- Cart
- Wishlist
- Checkout
- Orders
- Order Details
- Saved Reels
- Notifications
- Profile
- Settings

Innovator

- Dashboard
- Storefront
- Product Management
- Product Editor
- Inventory
- Orders
- Reels Manager
- Analytics
- Followers
- Verification Center
- Coupons
- Store Settings

Administrator

- Admin Dashboard
- User Management
- Innovator Verification
- Product Moderation
- Reel Moderation
- Innovation Verification
- Reports
- Categories

- CMS
- Support Tickets
- Audit Logs
- System Settings

22. Technology Stack

22.1 Technology Philosophy

The technology stack has been selected based on five principles:

- Scalability
- Performance
- Developer Productivity
- Community Support
- Long-Term Maintainability

The architecture should support rapid development while remaining flexible enough to scale from an MVP to an enterprise-grade marketplace.

22.2 Frontend Technology Stack

Technology	Purpose
Next.js (Latest)	React Framework, SSR, SEO
React 19	UI Library
TypeScript	Type Safety
Tailwind CSS	Utility-First Styling
Shadcn/UI	Modern Component Library
Radix UI	Accessible UI Primitives
Framer Motion	Animations & Microinteractions
TanStack Query	API State Management
Zustand	Global Client State
React Hook Form	Form Management
Zod	Form Validation

Technology	Purpose
Axios	HTTP Client
React Icons	Icons

22.3 Backend Technology Stack

Technology	Purpose
Node.js	Runtime Environment
Express.js	REST API Framework
TypeScript	Type Safety
JWT	Authentication
Bcrypt	Password Hashing
Multer	File Upload
Socket.IO	Real-Time Features
Cloudinary SDK	Image & Video Storage

22.4 Database

Primary Database:

MongoDB Atlas

Reason:

- Flexible document schema
 - Excellent support for nested product specifications
 - Easy horizontal scaling
 - Strong Node.js ecosystem
 - Generous free tier for development
-

22.5 Cloud Services

Media Storage

- Cloudinary

Deployment

Frontend

- Vercel

Backend

- Render (Initial)
 - Railway (Alternative)
-

22.6 Payment Gateway

India

- Razorpay

International

- Stripe

Future

- PayPal
 - Apple Pay
 - Google Pay
-

22.7 Development Tools

- Git
 - GitHub
 - VS Code
 - Postman
 - Figma
 - Docker (Future)
 - GitHub Actions (CI/CD)
-

23. High-Level System Architecture

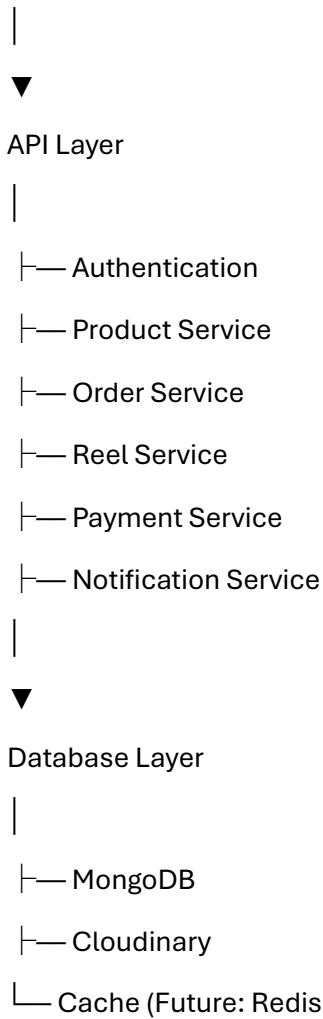
The platform shall follow a modular three-tier architecture.

Client Layer

|

|— Web Application (Next.js)

|— Future Mobile App



Architecture Principles

- Separation of Concerns
 - Modular Services
 - API-First Design
 - Stateless Backend
 - Cloud-Native Deployment
 - Scalable Infrastructure
-

24. Database Overview

The database shall support multiple interconnected entities.

Core Collections

User

Stores buyer, innovator, moderator, and administrator information.

Innovator Profile

Business information, verification status, followers, analytics.

Product

Stores complete product information including specifications, pricing, media, inventory, and verification details.

Product Category

Hierarchical category structure.

Innovation Reel

Stores reel metadata, product relationships, creator information, engagement statistics.

Order

Customer purchases and order lifecycle.

Cart

Temporary shopping session.

Wishlist

Saved products.

Review

Ratings and customer reviews.

Comment

Reel comments and discussions.

Notification

Platform notifications.

Coupon

Discount campaigns.

Payment

Payment transactions and reconciliation.

Support Ticket

Customer support interactions.

Audit Log

Administrative actions for traceability.

25. API Strategy

The platform shall expose RESTful APIs with consistent conventions.

API Versioning

/api/v1/

Future versions:

/api/v2/

/api/v3/

Authentication APIs

- Register
 - Login
 - Logout
 - Refresh Token
 - Forgot Password
 - Reset Password
 - Verify Email
-

Product APIs

- Create Product
 - Update Product
 - Delete Product
 - List Products
 - Product Details
 - Search Products
-

Reel APIs

- Upload Reel
 - Delete Reel
 - Feed
 - Like
 - Comment
 - Share
 - Save
-

Order APIs

- Create Order
 - Cancel Order
 - Track Order
 - Refund
-

Payment APIs

- Create Payment
 - Verify Payment
 - Payment History
-

Admin APIs

- User Management
- Seller Verification
- Product Approval

- Innovation Verification
 - Analytics
 - Reports
-

26. Assumptions & Constraints

Assumptions

- Users have internet access.
 - Sellers provide accurate product information.
 - Payment gateways remain available.
 - Cloud services maintain expected uptime.
 - Verification documents submitted are authentic.
-

Constraints

- Initial launch will target web browsers.
 - Mobile applications will follow in later phases.
 - AI features will be introduced after the MVP.
 - International logistics will be expanded progressively.
 - Verification is optional and subject to platform review.
-

27. Risk Analysis

Business Risks

- Low initial marketplace liquidity.
- Difficulty attracting both buyers and innovators simultaneously.
- Competition from established marketplaces.

Mitigation:

- Focus on innovation niche.
 - Build partnerships with universities, incubators, and startup ecosystems.
 - Feature verified innovators prominently.
-

Technical Risks

- Large media storage requirements.

- High traffic during promotional campaigns.
- Search performance at scale.

Mitigation:

- CDN-backed media delivery.
 - Scalable cloud infrastructure.
 - Search indexing and caching strategy.
-

Security Risks

- Fraudulent sellers.
- Counterfeit products.
- Malicious uploads.
- Account compromise.

Mitigation:

- Strong authentication.
 - Verification workflows.
 - File scanning.
 - Audit logging.
 - Rate limiting.
-

28. Development Roadmap

Phase 1 — Foundation

- Project setup
 - Design system
 - Authentication
 - Landing pages
-

Phase 2 — Marketplace

- Product management
- Categories
- Search
- Cart

- Checkout
-

Phase 3 — Innovation Feed

- Reel uploads
 - Feed algorithm
 - Engagement features
 - Product-tagged videos
-

Phase 4 — Seller Ecosystem

- Storefronts
 - Analytics
 - Verification
 - Inventory management
-

Phase 5 — Platform Administration

- Admin dashboard
 - Moderation
 - Reports
 - CMS
-

Phase 6 — Production Launch

- Performance optimization
 - Security hardening
 - SEO
 - Testing
 - Deployment
-

29. Future Scope

The platform architecture shall support future expansion without requiring major redesign.

Planned future capabilities include:

- AI-powered product recommendations.

- AI-assisted search.
- AI chatbot for customer support.
- Mobile applications (Android & iOS).
- Live shopping events.
- AR product visualization.
- B2B marketplace.
- Enterprise storefronts.
- Multi-language support.
- Multi-currency support.
- Affiliate marketing.
- Subscription-based innovator plans.
- Creator monetization tools.
- Logistics integrations.
- Patent database integration.
- AI-assisted Innovation Verification.
- AI-generated Innovation Scores.
- Innovation competitions and hackathon showcases.
- University and research institution partnerships.

30. Appendix

Glossary

- **Innovator** – A verified seller of innovative products.
- **Innovation Feed** – A short-form video discovery experience linked directly to products.
- **Innovation Verification** – An optional trust system that validates innovative products through supporting evidence.
- **Storefront** – A dedicated profile where an innovator manages products and branding.
- **Verification Badge** – A visual indicator showing successful completion of the Innovation Verification process.